

How the Particle Swarm Works

PSO controls swarm movements based on dynamic fish statistics: body/mouth size, velocity, energy, oxygen, and immunity. Individual trajectories are calculated through 12 weighted vectors navigating the 3D pond environment.

-  **cFood** 3x
Toward nearest food source when fullness < 100%
-  **cOxygen** 2x
Toward nearest oxygen bubble when level < 70%
-  **cSelfish** 1.5x
Away from nearest fish to maintain social distance
-  **cDisease** 2.5x
Away from identified nearest disease area
-  **cObstacle** 2x
Avoidance vectors for nearest physical barriers
-  **cRun** 5x
Evasion from nearest larger-mouthed fish
-  **cProbiotic** 1.5x
Toward probiotic when immunity < 100%
-  **cSocial** 1.5x
Toward swarm global optima and center mass
-  **cNH3** 4x
Away from nearest poison/ammonia area
-  **cParasite** 3x
Away from nearest parasite hotspot
-  **cRelief** 4x
Toward walls/obstacles to scrub off parasites
-  **cWander** 1x
Random stochastic search toward pond center